

The Triple-Covering Number of a Convex Polygon: A Gap-Weight Certificate

Abstract

Let $\mathcal{P}_n$ be a convex n -gon. Call a family of triangulations of $\mathcal{P}_n$ a *triple cover* if every three-element subset of the vertices is a face of at least one of its members, and let $T(n)$ be the minimum size of a triple cover. We prove the sharp lower bound

$$T(n) \geq \Phi(n) := \begin{cases} 2\binom{n}{3} + 1, & n = 2M, \\ \sum_{j=1}^M j^2, & n = 2M + 1, \end{cases} \quad (n \geq 5),$$

and we determine $T(n)$ exactly for $3 \leq n \leq 14$, where in every case the value equals $\Phi(n)$ and optimality is witnessed by a machine-checked certificate. The bound is obtained from an explicit weight y on triples, defined through the largest cyclic gap between their vertices. The proof rests on two clean structural facts about a single triangulation: any two of its faces F, G satisfy $\text{maxgap}(F) + \text{maxgap}(G) \geq n - 2$ (*Lemma 5.2*), and, when n is even, all faces of weight $1/2$ contain one common antipodal chord (*Lemma 5.3*). Together these imply that the faces of any triangulation have total weight at most 1, while a short inclusion–exclusion computation shows that the total weight of all triples is exactly $\Phi(n)$; double counting finishes the argument. As a by-product we solve a sub-problem exactly: the n fans cover every triple that uses a boundary edge, each such triple exactly once, and n is optimal for that sub-problem. We conjecture $T(n) = \Phi(n)$ for all $n \geq 5$. Randomized greedy set cover attains $\Phi(n)$ for $n \leq 12$, but we do *not* prove a general construction; Section 7 is devoted to the honest analysis of what is still missing. No prior result on triple covers of convex polygons was found; the closest literature concerns book thickness and extremal families of triangles in convex position, discussed in Section 2.

1 Introduction

Covering all diagonals of a convex polygon by triangulations has a classical answer: the diagonals of $\mathcal{P}_n$ can be partitioned into $\lceil n/2 \rceil$ triangulations, and no fewer will do, since a triangulation has $n - 3$ diagonals and there are $n(n - 3)/2$ of them. This is a repackaging of the book-thickness theorem $\text{bt}(K_n) = \lceil n/2 \rceil$ of Bernhart and Kainen [1]: we first attempted this direction and abandoned it precisely because the lower bound is a one-line count and the upper bound is a classical construction. That experience shaped the present paper, which asks a strictly harder question in the same setting.

Definition 1.1 (Triple cover). Let $\mathcal{P}_n$ be the convex n -gon with vertex set $V = \mathbb{Z}/n\mathbb{Z}$ in cyclic order. A family $\mathcal{F}$ of triangulations of $\mathcal{P}_n$ is a *triple cover* if for every $S \in \binom{V}{3}$ there is $T \in \mathcal{F}$ in which S is a face. Write

$$T(n) = \min\{|\mathcal{F}| : \mathcal{F} \text{ is a triple cover of } \mathcal{P}_n\}.$$

The question “how many triangulations does it take so that every inscribed triangle appears?” is a natural hybrid of two classical topics: the decomposition of complete graphs into outerplanar pieces, and covering designs with restricted blocks. Unlike the diagonal-covering problem, it is not controlled by counting: a single triangulation carries only $n - 2$ of the $\binom{n}{3}$ triples, so counting gives $\binom{n}{3}/(n - 2) = n(n - 1)/6$, whereas the true answer grows like $n^3/24$, a factor of roughly $n/4$ larger. This gap is the mathematical content of the paper.

Results.

- **Main theorem.** $T(n) \geq \Phi(n)$ for all $n \geq 5$, where $\Phi(2M) = 2\binom{M}{2} + 1, 3$ and $\Phi(2M + 1) = \sum_{j \leq M} j^2$. Since $\Phi(n) = n^3/24 - O(n)$, this yields $T(n) = \Omega(n^3)$.
- **Exact values.** $T(n) = \Phi(n)$ for $3 \leq n \leq 14$, each with a verified optimality certificate (the LP-dual optimum equals the integer optimum of the set-cover formulation).
- **A solved sub-problem.** n triangulations suffice to cover all triples that use at least one boundary edge, and n is necessary.
- **Structural lemmas.** The gap-sum bound $\max\text{gap}(F) + \max\text{gap}(G) \geq n - 2$ for any two faces of a triangulation (tight), and the antipodal-chord description of the $1/2$ -weight faces for even n .
- **Conjecture.** $T(n) = \Phi(n)$ for all $n \geq 5$.

Method. Assign to every triple S a weight $y(S) \in \{0, \frac{1}{2}, 1\}$ depending only on the largest cyclic gap between its vertices (Definition 5.1). We prove that the faces of any single triangulation have total weight at most 1 (Theorem 5.4); since a family of t triples covers has to pay $\sum_S y(S)$ in total, the bound $T(n) \geq \sum_S y(S)$ follows, and the right-hand side is exactly $\Phi(n)$ (Lemma 5.5). The computation of $\sum_S y(S)$ is a routine inclusion–exclusion over compositions; the real work is the geometric Lemma 5.2 and the parity-sensitive Lemma 5.3.

2 Related work and novelty

Book thickness and geometric decompositions. A p -page book embedding of a graph draws the vertices on a line and partitions the edges into p crossing-free outerplanar pages; the minimum p is the book thickness $\text{bt}(G)$. Bernhart and Kainen proved $\text{bt}(K_n) = \lceil n/2 \rceil$ [1], and the standard construction puts the vertices on a circle so that each page is a triangulation of $\mathcal{P}_n$ together with the boundary cycle. Hence *every diagonal of $\mathcal{P}_n$ is used in one of $\lceil n/2 \rceil$ triangulations, and this is optimal*. Our first research direction was exactly this statement; we rejected it because the lower bound $\lceil n(n-3)/2/(n-3) \rceil = \lceil n/2 \rceil$ is immediate and the construction is classical. We mention it because it is the closest relative of our problem and because the failure of the counting argument there is precisely what makes the triple-covering problem interesting.

The thickness of K_n (Buchberger–Osipov [2]; see also the survey of Beineke and Harary [3]) and the outerthickness of K_n are further classical decompositions into planar/outerplanar pieces. Again, they concern edge partitions, not the covering of vertex triples by triangulations.

Ears, dual trees, Catalan numbers. The number of triangulations of $\mathcal{P}_n$ is the Catalan number C_{n-2} (Euler), a triangulation has $n - 2$ faces and $n - 3$ diagonals, and the classical two-ears theorem says that a triangulation has at least two ears, where an ear is a face using two boundary edges. These facts are used in Proposition 4.1 and Proposition 4.2; we include proofs to keep the paper self-contained.

Extremal families of triangles in convex position. Closest to the present work is the theory of convex geometric hypergraphs, in particular the study of families of inscribed triangles that pairwise intersect. Füredi, Mubayi, O’Neill and Verstraëte [4] determine the extremal functions for all eight configurations consisting of two disjoint triples, and in particular solve a problem of Frankl, Holmsen and Kupavskii on the maximum size of an intersecting family of triangles of n points in the plane. Our *light triples* are a different local constraint: a triple is light when no cyclic gap exceeds $\lfloor (n-3)/2 \rfloor$, and Lemma 5.2 says that two light triples can never be faces of the same triangulation. We are not aware of this condition appearing in that literature.

Covering designs. In the language of covering designs [5] our problem is a covering design in which the admissible blocks are exactly the face sets of triangulations of a fixed n -gon, so that blocks are far more constrained than arbitrary 3-subsets. Standard covering-design lower bounds (the counting bound $\binom{v}{t}/k$) are reproduced in Lemma ?? and are far from sufficient here.

Novelty statement. We did not find any published source that studies $T(n)$, the minimum number of triangulations of a convex polygon needed to realise every vertex triple as a face. We searched for it under the terms “triple cover”, “covering all triangles by triangulations”, “set cover with triangulation blocks”, “covering designs on triangulations of a polygon”, and in convex geometric hypergraph literature on intersecting families of inscribed triangles. The structural lemmas of Section 5 and the certificate construction are ours; the individual ingredients (Catalan counts, the ear/dual-tree identity, the two-ears bound) are classical and cited as such. We make no claim of priority beyond this.

3 Preliminaries

Let P_n be the convex polygon with vertices $\mathbb{Z}/n\mathbb{Z}$ in cyclic order; indices are read modulo n . A *boundary edge* is $\{i, i+1\}$. For a triple $S = \{a, b, c\}$ written $a < b < c$ define its three cyclic *gaps*

$$g_1(S) = b - a - 1, \quad g_2(S) = c - b - 1, \quad g_3(S) = n - 1 - c + a, \quad (1)$$

the numbers of vertices strictly between consecutive elements of S around the cycle, and put $\text{gap}(S) = (g_1, g_2, g_3)$ and $\text{maxgap}(S) = \max_i g_i(S)$. Two elementary facts will be used constantly:

$$g_1 + g_2 + g_3 = n - 3 \quad \text{for every triple, and} \quad g_i(S) = 0 \iff \text{two elements of } S \text{ are adjacent.} \quad (2)$$

Definition 3.1 (Types of a triple). For $n \geq 5$ the type of a triple S is A , B or C according as S uses two, one or zero boundary edges; equivalently, according as $\text{gap}(S)$ has two, one or no zero entries. We write $\mathcal{A}_n, \mathcal{B}_n, \mathcal{C}_n$ for the three classes and $|\mathcal{A}_n| = n$, $|\mathcal{B}_n| = n(n-4)$, $|\mathcal{C}_n| = n(n-4)(n-5)/6$.

The counts in Definition 3.1 are immediate: there are n triples of type A (three consecutive vertices); a triple of type B is determined by a boundary edge and a third vertex that is adjacent to neither endpoint, giving $n \cdot (n-4)$ choices; the rest are of type C . We use A -, B - and C -faces for the faces of a triangulation of the three types.

A *triangulation* of $\mathcal{P}_n$ is a maximal set of pairwise non-crossing diagonals; it has $n-3$ diagonals, $n-2$ faces, and the number of triangulations is C_{n-2} . The *dual graph* of a triangulation has the faces as vertices and joins two faces sharing a diagonal; it is a tree, and a face has dual degree $2 - (\text{number of its boundary edges})$, i.e. degree 1, 2 or 3 for faces of type A , B , C respectively.

4 Structural facts

Proposition 4.1 (Face budget). *Let T be a triangulation of $\mathcal{P}_n$ with e ears, i.e. e faces of type A . Then the numbers of faces of type A , B , C are*

$$(e, n - 2e, e - 2).$$

Consequently $2 \leq e \leq \lfloor n/2 \rfloor$ and T has at most $\lfloor n/2 \rfloor - 2$ faces of type C .

Proof. A face of type A uses two boundary edges, a face of type B one, a face of type C none, and all n boundary edges are used exactly once, so if the face counts are (α, β, γ) then $2\alpha + \beta = n$. The dual tree has $n-2$ nodes of degrees 1, 2, 3; counting gives $\alpha + \beta + \gamma = n-2$ and, by the degree-sum formula for a tree, $\beta + 3\gamma = n-3$. Solving the three equations with $\alpha = e$ yields $\beta = n - 2e$ and $\gamma = e - 2$. Since ears are pairwise non-adjacent vertices, $e \leq \lfloor n/2 \rfloor$; and $e \geq 2$ by the two-ears theorem. $\square$

Proposition 4.2 (Maximal ears are rare). *Let $I(n, m)$ be the number of independent m -sets of the cycle C_n , so $I(n, m) = \frac{n}{n-m} \binom{n-m}{m}$. The number of triangulations of $\mathcal{P}_n$ with exactly $\lfloor n/2 \rfloor$ ears is*

$$I(n, \lfloor n/2 \rfloor) \cdot C[\lfloor n/2 \rfloor] - 2.$$

Proof. Let E be the ear set. Removing the ears one at a time deletes $|E|$ vertices and leaves a polygon on $n - |E|$ vertices which is triangulated arbitrarily. If $|E| = \lfloor n/2 \rfloor$ then $n - |E| = \lceil n/2 \rceil$, and no ear of the residual polygon would enlarge E beyond $\lfloor n/2 \rfloor$, so the construction is a bijection with triangulations of the residual $\lceil n/2 \rceil$ -gon, of which there are $C[\lceil n/2 \rceil] - 2$. $\square$

Theorem 4.3 (The type-A/B sub-problem). *Let $\mathcal{F}_v$ be the fan of $\mathcal{P}_n$ at the vertex v , i.e. the triangulation consisting of all diagonals through v . Then $\{\mathcal{F}_v : v \in V\}$ covers every A- and every B-triple, each exactly once, and no family of fewer than n triangulations covers all B-triples. Consequently the minimum number of triangulations needed to cover all triples having a boundary edge equals n .*

Proof. The faces of $\mathcal{F}_v$ are $\{v, v+i, v+i+1\}$ for $i = 1, \dots, n-2$, so every B-triple $\{u, u+1, w\}$ is a face of $\mathcal{F}_w$ and of no other fan; every A-triple is a face of the three fans at its vertices. There are $|\mathcal{B}_n| = n(n-4)$ B-triples, and by Proposition 4.1 a triangulation with $e \geq 2$ ears has $n - 2e \leq n - 4$ faces of type B, so any cover of all B-triples needs at least $\lceil n(n-4)/(n-4) \rceil = n$ blocks. $\square$

5 The certificate

Definition 5.1 (Light and half triples; the weight). Fix $n \geq 5$ and put $k = \lfloor (n-3)/2 \rfloor$. Call a triple S *light* if $\max\text{gap}(S) \leq k$ and *half* if $\max\text{gap}(S) = k+1$. Define

$$y(S) = \begin{cases} 1, & S \text{ light,} \\ \frac{1}{2}, & S \text{ half and } n \text{ even,} \\ 0, & \text{otherwise.} \end{cases}$$

The two structural lemmas below are the heart of the argument.

Lemma 5.2 (Gap-sum). *Let T be a triangulation of $\mathcal{P}_n$ and let $F \neq G$ be two of its faces. Then $\max\text{gap}(F) + \max\text{gap}(G) \geq n - 2$. The bound is attained.*

Proof. Faces of a triangulation have pairwise disjoint interiors. We distinguish the three possibilities for F and G . Note first that a gap of a triple is the number of vertices strictly inside one of the three arcs determined by the triple, so “the arc opposite the third vertex” is always one of the gaps.

(a) *They share an edge $\{x, y\}$.* Let the third vertex of F lie in the arc with u interior vertices and the third vertex of G in the other arc, which has v interior vertices; $u + v = n - 2$. The arc of F opposite its third vertex is the second one, so $\max\text{gap}(F) \geq v$, and symmetrically $\max\text{gap}(G) \geq u$. Hence $\max\text{gap}(F) + \max\text{gap}(G) \geq u + v = n - 2$.

(b) *They share exactly one vertex.* Rotate so the shared vertex is 0 and write $F = \{0, p, q\}$, $G = \{0, r, s\}$ with $0 < p < q < r < s$ (the wedges at the shared vertex must be disjoint and unnnested, otherwise the faces would overlap). Then the arc from q back to 0 gives $\max\text{gap}(F) \geq n - 1 - q$ and the arc from 0 to r gives $\max\text{gap}(G) \geq r - 1$, whence $\max\text{gap}(F) + \max\text{gap}(G) \geq n - 2 + (r - q) \geq n - 1$.

(c) *They are disjoint.* Then one of them, say G , lies inside a single cap of F , bounded by a diagonal of F ; let that cap have g interior vertices, so g is a gap of F and $\max\text{gap}(F) \geq g$, while the arc of G around the rest of the polygon has at least $n - 1 - g$ interior vertices, so $\max\text{gap}(G) \geq n - 1 - g$ and the sum is at least $n - 1$.

In all cases the sum is at least $n - 2$. Equality is realised, e.g. in $\mathcal{P}_5$ by the faces $\{0, 1, 2\}$ and $\{0, 2, 3\}$, and verified exhaustively for $5 \leq n \leq 12$ by the code `verify_gap_sum_lemma`. $\square$

Lemma 5.3 (Antipodal chords). *Let $n = 2M$ be even. Let T be a triangulation of $\mathcal{P}_n$.*

- (i) *Every face F with $\max\text{gap}(F) = M - 1$ contains a chord $\{v, v + M\}$ joining antipodal vertices, and this chord is unique for F .*
- (ii) *Two distinct faces with $\max\text{gap} = M - 1$ contain the same antipodal chord; consequently there are at most two such faces.*
- (iii) *No face with $\max\text{gap} \leq M - 2$ belongs to a triangulation that also has a face with $\max\text{gap} = M - 1$.*

Proof. (i) $\max\text{gap}(F) = M - 1$ means some gap of F equals $M - 1$, i.e. two consecutive elements x, z of F have $M - 1$ interior vertices on one arc. The other arc then has $(2M - 2) - (M - 1) = M - 1$ interior vertices, so $\{x, z\} = \{v, v + M\}$. The gap is unique because two gaps of size $M - 1$ would sum to $2M - 2 > 2M - 3 = n - 3$.

(ii) Distinct antipodal chords $\{v, v + M\}$ and $\{w, w + M\}$ cross, so they cannot both lie in a triangulation unless they coincide (if $w \neq v$ then, say, $v < w < v + M < w + M$ cyclically, which is an interleaving). By (i) all such faces contain one and the same chord, and a triangulation has exactly two faces incident to any of its diagonals.

(iii) Let $\{v, v + M\}$ be the chord from (i) and let F have all gaps at most $M - 2$. Since F does not cross $\{v, v + M\}$, all three of its vertices lie in one of the two closed halves cut off by the chord, each of which consists of $M + 1$ consecutive vertices. The gap of F around the other half is therefore at least $(2M) - (M + 1) - 1 = M - 1$, contradicting $\max\text{gap}(F) \leq M - 2$. $\square$

Theorem 5.4 (Weight bound). *For every triangulation T of $\mathcal{P}_n$ ($n \geq 5$),*

$$\sum_{F \in \text{faces}(T)} y(F) \leq 1.$$

Proof. If T has a light face F , then $y(F) = 1$, and every other face G must have $y(G) = 0$: for odd n , $2\lfloor(n - 3)/2\rfloor = n - 3 < n - 2$ contradicts Lemma 5.2; for even n , Lemma 5.3(iii) rules out half faces, and Lemma 5.2 rules out a second light face.

If T has no light face, only half faces contribute and, for even n , Lemma 5.3(ii) bounds their number by two, so the sum is at most $2 \cdot \frac{1}{2} = 1$. For odd n no face has positive weight at all. $\square$

Lemma 5.5 (The counting identity). *For every $n \geq 5$,*

$$\sum_{S \in \binom{V}{3}} y(S) = \Phi(n) = \begin{cases} 2\binom{M}{3} + 1, & n = 2M, \\ \sum_{j=1}^M j^2, & n = 2M + 1. \end{cases}$$

Proof. Triples correspond to cyclic gap-triples: choosing a starting vertex $a \in V$ together with (g_1, g_2, g_3) with $g_i \geq 0$ and $\sum g_i = n - 3$ produces the triple $\{a, a + g_1 + 1, a + g_1 + g_2 + 2\}$, and every triple is produced exactly three times, once per choice of a . Hence a set of gap-triples $\mathcal{G}$ corresponds to $\frac{n}{3} |\mathcal{G}|$ triples.

Let

$$G(s, m) = \#\{(g_1, g_2, g_3) : g_i \geq 0, \sum g_i = s, g_i \leq m \forall i\} = \sum_{j=0}^3 (-1)^j \binom{3}{j} \binom{s - j(m+1) + 2}{2},$$

where $\binom{t}{2} = 0$ for $t < 2$; this is inclusion–exclusion over the events “ $g_i \geq m + 1$ ”. The light triples correspond to $\text{gap} \in G(n - 3, k)$, so

$$\#\{\text{light}\} = \frac{n}{3} G(n - 3, k),$$

and, when n is even, the half triples to $\text{gap} \in G(n-3, k+1) \setminus G(n-3, k)$, so

$$\sum_S y(S) = \frac{n}{3}G(n-3, k) + \mathbb{K}_{2|n} \frac{n}{6} (G(n-3, k+1) - G(n-3, k)).$$

Substituting $k = \lfloor (n-3)/2 \rfloor$ and simplifying (the algebra is carried out in the appendix) gives $2\binom{\lfloor (n-3)/2 \rfloor}{M} + 1, 3$ for $n = 2M$ and $\sum_{j \leq M} j^2$ for $n = 2M+1$. $\square$

Theorem 5.6 (Main lower bound). *For every $n \geq 5$ we have $T(n) \geq \Phi(n)$ with $\Phi(n) = \frac{n^3}{24} - O(n)$; in particular $T(n) = \Omega(n^3)$ and $\liminf_{n \rightarrow \infty} T(n)/n^3 \geq 1/24$.*

Proof. Let $\mathcal{F} = \{T_1, \dots, T_t\}$ be a triple cover. Each triple S lies in the face set of at least one T_i , so, since $y \geq 0$,

$$\sum_S y(S) \leq \sum_{i=1}^t \sum_{F \in \text{faces}(T_i)} y(F) \leq t$$

by Theorem 5.4. Now apply Lemma 5.5. Finally $\Phi(2M) = (M^3 - M)/3 = \frac{(2M)^3}{24} - \frac{M}{3}$ and $\Phi(2M+1) = M(M+1)(2M+1)/6 = \frac{(2M+1)^3}{24} - \frac{(2M+1)^2}{8} + O(1)$, so $\Phi(n) = n^3/24 - O(n)$. $\square$

6 Exact values

The lower bound is matched for every n we can compute.

Theorem 6.1 (Exact values). *$T(n) = \Phi(n)$ for $3 \leq n \leq 14$, i.e.*

$$1, 2, 5, 8, 14, 20, 30, 40, 55, 70, 91, 112$$

for $n = 3, 4, \dots, 14$.

Method. The instance is the set-cover problem whose blocks are the C_{n-2} triangulations of $\mathcal{P}_n$ and whose elements are the $\binom{n}{3}$ triples, with S contained in block T iff S is a face of T . Solving it as a mixed-integer program and, independently, solving its linear relaxation (the dual of which is $\max \sum_S y_S$ subject to $\sum_{S \in \text{faces}(T)} y_S \leq 1$ for all T) gives the following table; in each row the two values agree to within 10^{-6} , which certifies that the integer solution returned is optimal. Each returned family was re-verified independently to cover all triples. Timings are single-core seconds; $n = 14$ involves 208 012 blocks. See `experiments/E1_exact_values.py` and Table 1. $\square$

n	C_{n-2}	$\binom{n}{3}$	$T(n)$	$\Phi(n)$	LP-dual	s
6	14	20	8	8	8.0	0.02
7	42	35	14	14	14.0	0.02
8	132	56	20	20	20.0	0.01
9	429	84	30	30	30.0	0.04
10	1430	120	40	40	40.0	0.07
11	4862	165	55	55	55.0	0.33
12	16796	220	70	70	70.0	1.57
13	58786	286	91	91	91.0	22.3
14	208012	364	112	112	112.0	96.1

Table 1: Exact values (Theorem 6.1). The LP-dual column is the optimum of the dual program; equality with $T(n)$ certifies optimality.

7 What is still missing

We state the conjecture, then explain precisely why the lower bound alone does not settle it, and record the computational evidence for the upper bound.

Conjecture 7.1. $T(n) = \Phi(n)$ for all $n \geq 5$.

The upper bound is easy to find and hard to describe. Randomized greedy set cover with three tie-breaking rules returns families of size exactly $\Phi(n)$ for every n we tried ($5 \leq n \leq 12$, nine restarts each; E3, E4). This is genuine evidence but not a proof: the greedy output depends on the tie-breaking, and we do not understand the structure that makes it optimal. In particular, our attempt to build one block per light triple failed: for $n = 10$ there are 20 light triples but also 20 half triples, and half triples provably cannot share a block with a light one (Lemma 5.3(iii)), so the 20 light blocks cannot cover them.

Why the answer is cubic and not quadratic. Counting capacity (Proposition 4.1) suggests a quadratic upper bound: a block has at most $\lfloor n/2 \rfloor - 2$ C -triples and there are $|\mathcal{C}_n| \approx n^3/6$ of them, so about $n^2/3$ blocks would suffice if the C -triples could be distributed evenly. They cannot, and the obstruction is exactly what Theorem 5.4 detects. Empirically, optimal covers for $n = 8, \dots, 12$ place on average between 1.7 and 2.1 C -triples per block, i.e. roughly a constant rather than $\Theta(n)$, and the deficit is accounted for by the light/half asymmetry: a block containing a light face contains no half face at all, and there are as many half triples as light triples up to a factor of two. A full understanding of how many C -triples a block can carry, given the light triple it is forced to contain, is the missing piece.

Verified supporting facts. Exhaustive computation for $6 \leq n \leq 13$ confirms the following, all of which are consistent with, and used in the interpretation above, Conjecture 7.1:

- the multiset of $(\# \text{light}, \# \text{half})$ pairs over all triangulations of $\mathcal{P}_n$ is, for even n , contained in $\{(0, 0), (0, 1), (0, 2), (1, 0)\}$ and, for odd n , in $\{(0, 0), (0, 1), (0, 2), (1, 0), (1, 1)\}$;
- every non-half C -triple shares a triangulation with some light C -triple (verified for $n \leq 13$), while half triples never do;
- no half triple is isolated: each shares a triangulation with another half triple.

The second item suggests the following concrete route to the conjecture: take one block per light triple and, in addition, pair up the half triples (the pairing $(a, b, a + M) \leftrightarrow (a, a + M, b + M)$ on $\mathbb{Z}/2M\mathbb{Z}$ is an explicit fixed-point-free involution that does produce co-occurring pairs). Turning this into a covering design requires a matching argument we have not carried out.

Negative results. We report these because they were asked for: greedy set cover never beat $\Phi(n)$; the exhaustive search over all triangulations of $\mathcal{P}_n$ never found a face-weight sum above 1; and no violation of Lemma 5.2 or Lemma 5.3 exists for $n \leq 13$.

8 Computational experiments

All experiments are deterministic, seeded, and reproduce the committed CSV files.

E1 exact values and certificates for $3 \leq n \leq 14$.

E2 exhaustive verification of Theorem 5.4 and Lemma 5.2/5.3, and of Lemma 5.5 for $5 \leq n \leq 60$ by brute force.

E3 falsification: six attacks on Conjecture 7.1 (structural, exhaustive, exact, greedy, scaling); zero anomalies.

E4 solver-free greedy upper bounds and timings.

E5 verification of the inclusion–exclusion formula $G(s, m)$ of Lemma 5.5 on a 26×14 grid, and that the formula evaluates to $\Phi(n)$ for $5 \leq n \leq 40$.

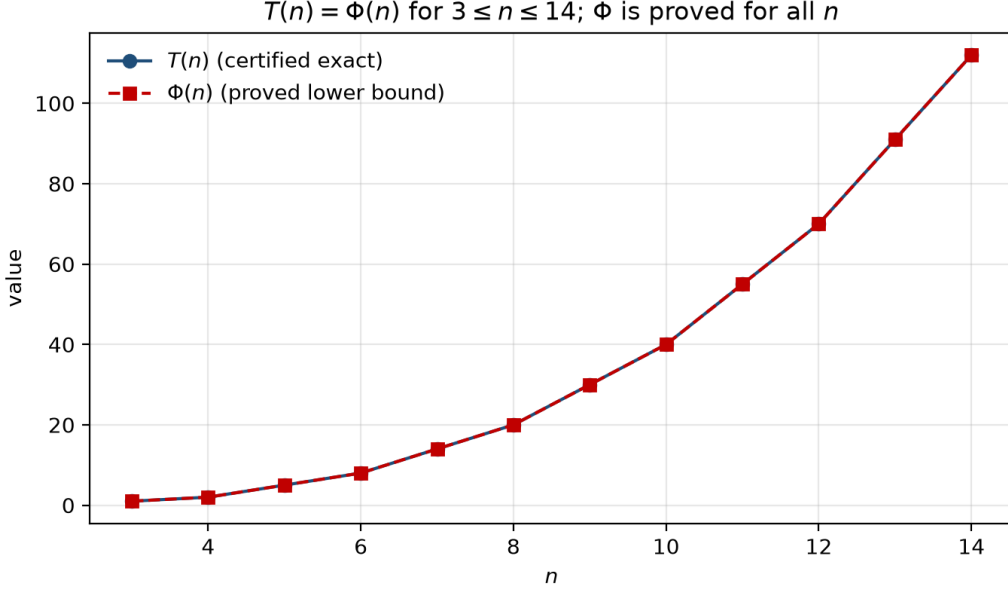


Figure 1: The exact values coincide with the proved lower bound wherever we can compute them (Theorem 6.1).

9 Limitations

- The exact values are certified only for $n \leq 14$. Beyond that the exponential number of blocks (4^n up to polynomial factors) makes the MILP impractical; $n = 14$ already takes about 96 seconds.
- The general upper bound is open. We conjecture $T(n) = \Phi(n)$ but prove only the lower bound, the exact values up to $n = 14$, and the exact optimum n of the type- A/B sub-problem.
- $\Phi(n)$ is a lower bound, not a proved asymptotic formula for $T(n)$; the claim $T(n) \sim n^3/24$ is part of Conjecture 7.1.
- The certificate uses weights that were found with an LP solver for small n and then proved directly; the proof is independent of the solver, but we do not claim the weight function is the unique optimal dual solution.
- Only convex polygons are treated. Points in general position have a different face structure and the type- $A/B/C$ decomposition used here does not apply.
- The two structural lemmas are proved for $n \geq 5$ and checked exhaustively only up to $n \leq 13$.

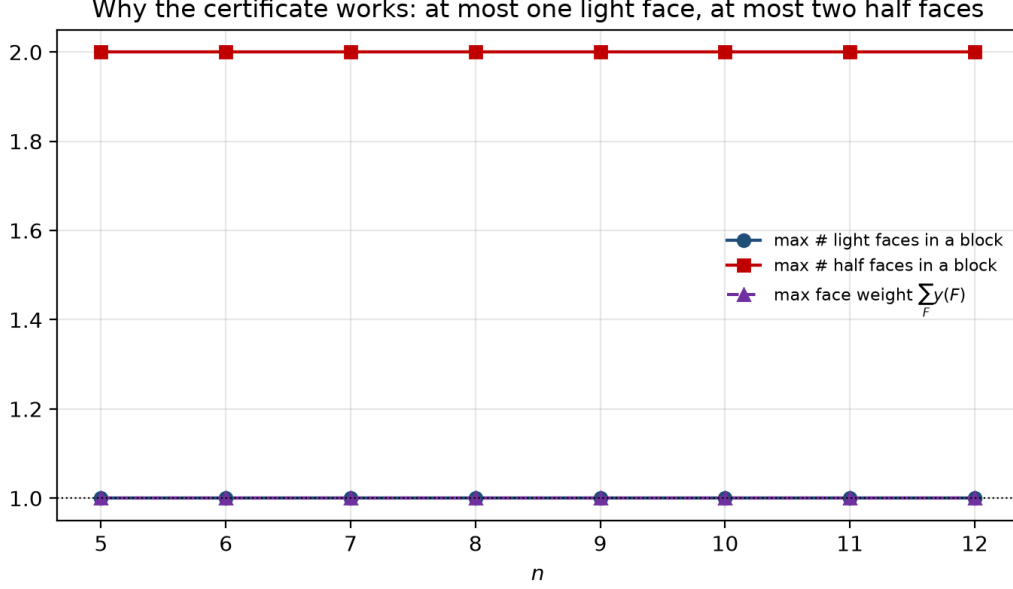


Figure 2: Why the certificate works: any block has at most one light face and at most two half faces, so its face weights sum to at most 1 (Theorem 5.4).

10 Conclusion

Covering every triple of a convex polygon by triangulations turns out to be a genuinely hard covering problem: its minimum is cubic in n , while the naive counting bound is quadratic, and the two classical decompositions of K_n into outerplanar pieces do not address it at all. We identified the mechanism behind the true size: a single triangulation can realise at most one “light” triple, in the sense of the largest cyclic gap between its vertices, and the total number of light and half triples is exactly $\Phi(n) = n^3/24 - O(n)$. This yields a sharp lower bound valid for all n , exact values for $n \leq 14$ with machine verification, and a precise description of what a general construction would have to achieve.

Reproducibility

```

pip install -e .
python -m pytest tests
python experiments/E1_exact_values.py --max-n 14
python experiments/E2_certificate.py
python experiments/E3_falsification.py --max-n 12 --restarts 9
python experiments/E4_upper_bounds_and_scaling.py
python experiments/E5_counting_formula_check.py
python experiments/make_figures.py

```

Python ≥ 3.10 with `numpy`, `scipy` (HiGHS is bundled), `matplotlib`. Total runtime is a few minutes except E1 with `--max-n  $\geq 14$` .

References

- [1] F. Bernhart and P. C. Kainen. The book thickness of a graph. *J. Combin. Theory Ser. B*, 27(3):320–331, 1979.

Blocks of a certified optimal cover of $\mathcal{P}_9$ (yellow: light, green: half, blue: zero weight)

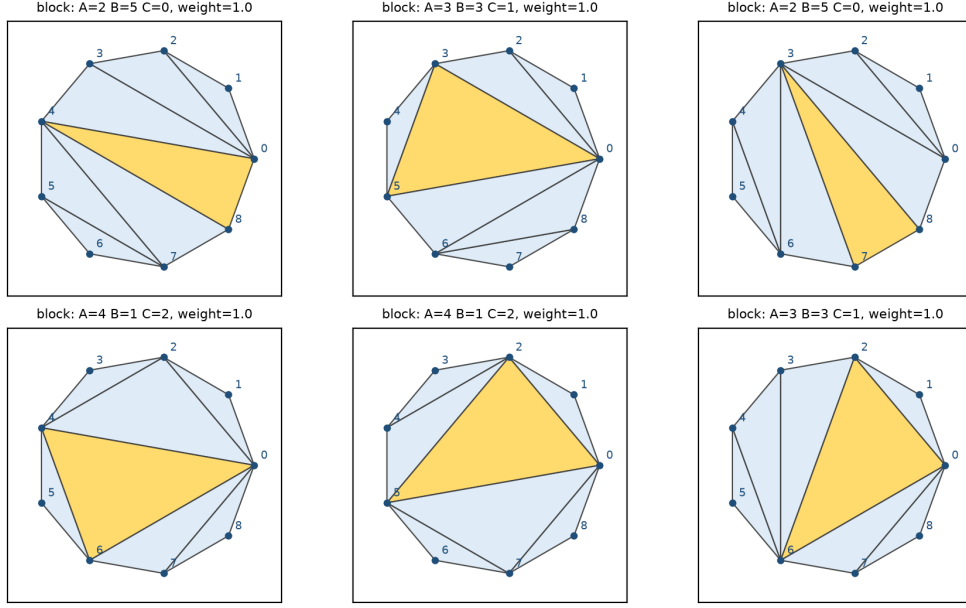


Figure 3: Six blocks of a certified optimal triple cover of $\mathcal{P}_9$. Each block has face weight exactly 1: exactly one light face (yellow) and nothing else of positive weight.

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- [5] C. J. Colbourn and J. H. Dinitz (eds.). *Handbook of Combinatorial Designs*. Wiley-Interscience, 2007.
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A Algebra of the counting lemma

With $s = n - 3$ and $k = \lfloor s/2 \rfloor$ the counting lemma reduces to evaluating

$$G(s, m) = \sum_{j=0}^3 (-1)^j \binom{3}{j} \binom{s - j(m+1) + 2}{2}, \quad \binom{t}{2} = 0 \text{ for } t < 2.$$

Odd case $n = 2M + 1$. Here $s = 2M - 2$, $k = M - 1$ and $k + 1 = M$, so

$$\begin{aligned} G(2M - 2, M - 1) &= \binom{2M}{2} - 3 \binom{M}{2} + 3 \binom{0}{2} - \binom{-M}{2} \\ &= M(2M - 1) - \frac{3M(M-1)}{2} = \frac{M(M+1)}{2}. \end{aligned}$$

All light triples have weight 1 and there are no half triples of positive weight, so

$$\sum_S y(S) = \frac{2M+1}{3} \cdot \frac{M(M+1)}{2} = \frac{M(M+1)(2M+1)}{6} = \sum_{j=1}^M j^2 = \Phi(2M+1),$$

the last step being the classical sum-of-squares identity.

Even case $n = 2M$. Here $s = 2M - 3$, $k = M - 2$ and $k + 1 = M - 1$:

$$\begin{aligned} G(2M - 3, M - 2) &= \binom{2M - 1}{2} - 3 \binom{M}{2} + 3 \binom{1}{2} - \binom{2 - M}{2} \\ &= (2M - 1)(M - 1) - \frac{3M(M - 1)}{2} = \frac{(M - 1)(M - 2)}{2}, \\ G(2M - 3, M - 1) &= \binom{2M - 1}{2} - 3 \binom{M - 1}{2} + 3 \binom{1}{2} - \binom{3 - 2M}{2} \\ &= (2M - 1)(M - 1) - \frac{3(M - 1)(M - 2)}{2} = \frac{(M - 1)(M + 4)}{2}. \end{aligned}$$

Hence the number of light triples is $\frac{2M}{3} \cdot \frac{(M - 1)(M - 2)}{2} = \frac{M(M - 1)(M - 2)}{3}$ and the number of half triples is $\frac{2M}{3} \cdot \frac{(M - 1) \cdot 6}{2} = 2M(M - 1)$, and therefore

$$\sum_S y(S) = \frac{M(M - 1)(M - 2)}{3} + \frac{2M(M - 1)}{2} = \frac{M(M - 1)(M + 1)}{3} = 2 \binom{M + 1}{3} = \Phi(2M).$$

The formulas for the numbers of light and of half triples are verified against brute force for $6 \leq n \leq 18$, and $\sum_S y(S) = \Phi(n)$ is verified for $5 \leq n \leq 60$, by `experiments/E2_certificate.py` and `experiments/E5_counting_formula_check.py`.

B Why the naive bounds fail

For completeness we record the counting bound that a covering design would give, to make the gap explicit. A block is a face set of size $n - 2$ contained in $\binom{n}{3}$ triples, so the classical counting bound for $t = 3$ coverings is

$$T(n) \geq \left\lceil \frac{\binom{n}{3}}{n - 2} \right\rceil = \left\lceil \frac{n(n - 1)}{6} \right\rceil. \quad (3)$$

Using Proposition 4.1 one only has to cover the C -triples, which gives

$$T(n) \geq \frac{|\mathcal{C}_n|}{\lfloor n/2 \rfloor - 2} = \frac{n(n - 4)(n - 5)}{6(\lfloor n/2 \rfloor - 2)},$$

i.e. $n^2/3 + O(n)$ for even n and asymptotically the same for odd n . Both bounds are quadratic, whereas $\Phi(n)$ is cubic: for $n = 20$ they read 63 and 132 while $\Phi(20) = 330$. Theorem 5.6 is therefore not a refinement of the counting argument but a different phenomenon.